

LLM Zero-Shot Distress Lexicon: Research Protocol

Airtight Methodology, Sub-Category Architecture, and Validation Pipeline

1. Executive Summary & 3-Tier Representation Framework

This research evaluates the marginal predictive value of unstructured loan descriptions across three levels of text representation. The LLM serves strictly as a **Zero-Shot Domain Annotator** to build an interpretable, domain-specific *Consumer Distress Lexicon*, avoiding any arbitrary target word counts.

Representation Level	Methodology	Feature Engineering	Academic Rationale
Level 1: Lexicon	Zero-Shot LLM Distress Lexicon & Sub-Indices	Aggregate scores (Count, Density, Flag) + Sub-category distress counts.	Highly interpretable, domain-specific rule baseline capturing categorized financial distress without black-box opacity.
Level 2: Statistical	TF-IDF (Unigrams & Bigrams)	Corpus-wide weighted n-gram matrix.	Standard statistical baseline measuring vocabulary-wide term importance without syntactic awareness.
Level 3: Deep NLP	FinBERT Contextual Embeddings	Dense 768-dimensional sentence vectors.	State-of-the-art transformer baseline capturing bidirectional sentence structure, context, and negation.

2. Step-by-Step Methodological Workflow

- Strict Vocabulary Extraction & Freezing (No Data Leakage):** Extract the top N frequent unigrams and bigrams exclusively from the **initial baseline training window** prior to running the walk-forward expanding cross-validation. Apply minimum frequency pruning ($freq > 15$). The extracted lexicon is **frozen** and applied unchanged across all subsequent walk-forward folds.
- Zero-Shot LLM Annotation via Structured Rubric:** Submit the frozen n-gram vocabulary to the LLM ($T = 0.0$) to annotate each term into a structured Distress category or Neutral. The number of active risk terms is determined purely by the rubric criteria (an empirical outcome, not a pre-set quota).
- Human-in-the-Loop Validation:** Draw a random stratified sample (150–200 terms) across annotated classes. Conduct blind human expert annotation and report inter-rater reliability (Cohen's κ and Precision) to establish construct validity.
- Primary & Secondary Feature Engineering on Loan Text:**
 - Primary Tabular Features (Parsimonious & Interpretable):**
 - Count** risk_word_count: Total distress terms identified in description.
 - Density** risk_word_density: $Count(Distress\ Words) / Total\ Word\ Count$.
 - Flag** risk_word_flag: Binary indicator (1 if ≥ 1 distress term present, 0 otherwise).
 - Sub-Category Distress Features (Domain Insights):** Generate aggregated count features for distinct distress dimensions:
 - distress_debt_pressure_count (e.g., compounding interest, maxed limits, juggling payments)
 - distress_liquidity_shortage_count (e.g., behind on rent, urgent cash flow, overdraft)
 - distress_medical_life_shock_count (e.g., illness, unexpected surgery, family emergency)
 - distress_refinancing_burden_count (e.g., predatory consolidation, payday loans)

- **Secondary / Robustness Analysis (Optional):** Individual binary dummies for specific terms evaluated only in an ablation subsection.

5. **Downstream Walk-Forward Evaluation:** Train default prediction algorithms (e.g., XGBoost, LightGBM, Regularized Logistic Regression) combining structured credit tabular variables with text features, evaluating out-of-fold AUC-ROC, Log-Loss, and fairness metrics across geographic/demographic slices.

3. Formal Zero-Shot Prompt Rubric & Multi-Class Categories

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SYSTEM: You are an expert quantitative credit risk analyst. Your objective is to annotate a vocabulary list extracted from peer-to-peer personal loan descriptions. Classify each term into EXACTLY ONE of the following categories: - DEBT_PRESSURE: Terms signaling high debt load, maxed credit lines, mounting interest, or debt-spiral traps. - LIQUIDITY_SHORTAGE: Terms signaling immediate cash-flow deficit, inability to cover basic living expenses, or being past due. - SHOCK_EMERGENCY: Terms signaling unexpected exogenous financial distress (medical emergencies, sudden job loss, urgent repairs). - HIGH_RISK_REFINANCING: Terms explicitly indicating payoff of predatory/ payday debt or high-interest short-term borrowing. - NEUTRAL_BENIGN: Terms describing routine loan logistics, stable employment, positive sentiment, planned life events, or general conversational prose. Output format (CSV): token,category,is_distress(0/1),rationale
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4. Scientific Reproducibility & Audit Tracking

Audit Item	Specific Requirement to Log & Commit to Repo
Model Specification	Exact model release identifier (e.g., c1aude-3-5-sonnet-20241022), API client version, and execution timestamp.
Inference Parameters	Temperature fixed at 0.0 (strict determinism), top-\$p = 1.0\$, and max token allocation.
Prompt Artifacts	Exact verbatim system and user prompts stored in version control (e.g., prompts / zero_shot_lexicon_rubric_v2.json).
Raw & Curated Tables	Complete raw API output CSV alongside the final filtered sub-categorized lexicon table.
Validation Metrics	Confusion matrix, Cohen's κ , and Precision scores comparing LLM labels vs. human expert sample.

5. Academic Defense Strategy for Advisor & Committee

Core Committee Argument: We replace out-of-domain corporate accounting lexicons with a frozen, zero-shot consumer distress dictionary whose vocabulary volume is an empirical outcome of strict risk rubrics, validated with human-in-the-loop agreement and leak-free temporal evaluation.

- **Loughran-McDonald Incompatibility:** L&M was calibrated on formal SEC 10-K filings and fails to transfer to informal borrower prose. Demonstrating this domain gap justifies the need for a consumer-specific lexicon.
- **Empirical Vocabulary Size:** Rather than forcing an arbitrary quota, the final lexicon size is a natural result of the classification rubric. Natural sparsity in borrower self-reporting is a well-documented phenomenon.
- **Methodological Rigor via Freezing & Human Audit:** Freezing the lexicon on the initial training window prevents temporal data leakage, while human sample validation provides verifiable construct validity.